

Indraprastha Institute of Information Technology, Delhi

IIIT-Delhi - Department of Computer Science & Engineering

Discrete Mathematics - Course Syllabus

Course Resource Package

Course Code: CSE111 | Course: Discrete Mathematics

1. Logic & Proofs: Propositional logic, quantifiers, rules of inference, mathematical induction.
2. Sets, Relations & Functions: Equivalence relations, poset, lattices, growth of functions.
3. Combinatorics: Permutations, combinations, generating functions, recurrence relations.
4. Graph Theory: Isomorphism, Euler/Hamilton paths, planar graphs, graph coloring.
5. Group Theory: Groups, subgroups, Lagrange's Theorem, cyclic groups.